Information & Management 54 (2017) 479–490

Contents lists available at ScienceDirect

Information & Management

journal homepage: www.elsevier.com/locate/im

Virtual team effectiveness: The role of knowledge sharing and trust

Mohammad Alsharo[a,*], Dawn Gregg[b,*], Ronald Ramirez[b]

[a] Information Systems, Al Albayt University, P.O. BOX 130040, Mafraq 25113, Jordan
[b] Information Systems, University of Colorado Denver, 1380 Lawrence Street, Denver, CO 80204, USA

ARTICLE INFO

Article history:
Received 21 February 2016
Received in revised form 3 October 2016
Accepted 31 October 2016
Available online 5 November 2016

Keywords:
Virtual team
Knowledge sharing
Trust
Collaboration
Team effectiveness
Knowledge based theory of the firm

ABSTRACT

Organizations utilize virtual teams to gather experts who collaborate online to accomplish organizational tasks. The virtual nature of these teams creates challenges to effective collaboration and team outcomes. This research addresses the social effects of knowledge sharing on virtual teams. We propose a conceptual model which hypothesizes a relationship between knowledge sharing, trust, collaboration, and team effectiveness in virtual team settings. The findings suggest that knowledge sharing positively influences trust and collaboration among virtual team members. The findings also suggest that while trust positively influences virtual team collaboration, it does not have a significant direct effect on team effectiveness.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

Teams are used by firms to organize employee resources to support the completion of complex and non-routine tasks. If designed and managed properly, teams can contribute to organizational value and competitive advantage [3]. Organizations have shifted to team-based work over the past several decades, with advances in information and communication technologies (ICT) enabling and further motivating this shift. Team design has moved beyond a configuration of same-location employees to now include members from different geographic locations and from different organizations [10]. The result is the creation of virtual teams with new types of work patterns, decision making styles, and relationships. Together, these changes pose unique challenges to theory, practice, and research on team effectiveness [10,79,18]. Especially as traditional team effectiveness factors may not apply or be less effective in a virtual team setting [10].

In a knowledge-based view of the firm [60], knowledge is seen as an organizational asset that can be used to create and sustain a competitive advantage for both organizations and individuals. Organizations, as knowledge-based systems, must recognize the significance of knowledge as a valuable asset that has the potential to impact prosperity, maximize economic value, and improve effectiveness [28,2]. As knowledge is distributed among organizational employees and other non-affiliated individuals, the use of teams can facilitate the sharing and integration of knowledge to improve performance and outcomes.

Organizations establish teams to bring together individuals with the necessary expertise and skills to collaborate on organizational tasks [37]. The advantage of the team structure is that it integrates the knowledge that is distributed among team members, which facilitates synergies and the achievement of more effective outcomes [51]. However, knowledge sharing is dependent on the willingness of individual team members to share the unique knowledge they possess [11] and unfortunately, a common perception is that sharing knowledge may lead to a loss of knowledge ownership and related loss of power. This can lead to the hoarding of knowledge by employees and a reluctance to sharing [43]. As knowledge sharing and idea exchange is crucial for virtual team collaboration, an effective virtual team requires its members to put the success of the team ahead of their tendency to hoard knowledge.

Trust is a key factor for addressing these challenges and is central element that can affect team effectiveness and success [40,88,36]. However, building trust among team members in a virtual team setting is a complex task. Virtual team members are unable to observe physical behaviors which traditional face-to-face team members rely upon to establish and sustain trust. Virtual team members are reliant upon on different behaviors, unique to virtual settings, to assess trustworthiness and compensate for the lack of physically observed behaviors. We see knowledge sharing as a key behavior that virtual team members can observe and rely upon to build trust within their team.

* Corresponding authors.
E-mail addresses: mohammad.alsharo@aabu.edu.jo (M. Alsharo), dawn.gregg@ucdenver.edu (D. Gregg), ronald.ramirez@ucdenver.edu (R. Ramirez).

http://dx.doi.org/10.1016/j.im.2016.10.005
0378-7206/© 2016 Elsevier B.V. All rights reserved.

In this research we investigate the role of knowledge sharing and trust in enabling virtual team as well as their impact on team effectiveness. Using the Knowledge Based Theory of The Firm, Social Capital Theory, and Social Exchange Theory, we propose a novel conceptual model which explains the relationship between knowledge sharing, trust, and team effectiveness, taking into account the unique characteristics of virtual teams. We test this model using data generated by a survey of virtual members working on a random set of projects in an organizational setting. The results of structural equation modeling analyses indicate that knowledge sharing positively influences trust and collaboration among virtual team members. The findings also suggest that trust positively influences virtual team collaboration, but, does not have a direct effect on team effectiveness, instead the impact of trust on team effectiveness is fully mediated by collaboration.

Our research extends the Knowledge Based Theory of the Firm by improving our understanding of how knowledge sharing impacts trust, collaboration, and virtual team effectiveness. It is one of the first studies to examine the role knowledge sharing plays in building trust and collaboration in virtual team settings. Our research also contributes to a recent call for the investigation into socio-emotional processes in virtual settings to better understand what contributes to team effectiveness [10,18,70].

2. Background

In the 1990s, organizations typically established virtual teams to work on temporary projects with short life cycles. These teams were mainly established based on a need to quickly gather necessary knowledge and expertise to solve complex or non-routine problems [78]. The temporary nature of early virtual teams afforded limited opportunities to form social relationships [40]. Today, however, organizations are increasingly establishing virtual teams to work on ordinary routine tasks. In addition, organizations are allowing employees to work virtually from the location of their preference. The resulting mix of teams in an organization, some completely distributed while others a mix of co-located and distributed, is deemed as the degree of team virtualness [31].

Due to the powerful combined intellectual capital of team members who benefit from each other's knowledge and expertise, virtual teams have the potential to generate innovative and valuable results [70]. As such, investigating virtual teams is essential for learning how firms can deploy resources and perform tasks for the maximization of performance, especially when compared to traditional face-to-face team settings [71].

Beyond the technology used by teams for communication and collaboration, a major influence on virtual team performance is the social interaction and social capital among team members [71,18,70]. According to Social Presence Theory, "*communication is effective if the communication medium has the appropriate social presence required for the level of interpersonal involvement required for a task*" [75]. Organizations are social systems of individuals who are required to work collaboratively on accomplishing a common task [3]. Collaboration is an essential part of team work and an effective collaboration leads to an effective team outcome [5]. Virtual team members are chosen and assigned to tasks based on the need for their unique knowledge and expertise. However, recruiting expert members does not guarantee effective collaboration [35]. Adopting elements of Aram and Morgan [5], we define collaboration in an organizational setting as " . . . the presence of mutual influence between persons, open and direct communication and conflict resolution, and support for innovation and experimentation . . . " (Page 1). Team effectiveness has to do with group-produced outputs, and to collaborate effectively the knowledge that is distributed among team members must be properly and adequately integrated [30]. Thus, we propose that for virtual teams to collaborate effectively, team members must establish open communication and effectively share knowledge among each other.

Adopting elements of Alavi and Leidner [2], we define knowledge as, " . . . a justified belief that increases an entity's capacity for taking effective action . . . " (Page 14). In relation to a firm, knowledge can be possessed by employees or by the organization as a whole (the entity) and is used, through skill or capability, to complete individual or organizational tasks (the action). Knowledge is an organizational asset that comes in tacit and explicit forms, and should be treated as a significant organizational resource that can contribute to an organization's competitive advantage [60,2].

The origination of knowledge takes place at both the individual and group levels [60]; individuals create knowledge independently whereas teams create a social knowledge through the interaction and collaboration of team members [2]. Transforming individual knowledge into social knowledge is a complex task as it depends on employee willingness to share and contribute individual knowledge [11] given the tendency to hoard the knowledge for preservation of value within an organization [27,11]. The reluctance of employees to share knowledge has the potential to hinder team collaboration and rresult in missed objectives [81]. Virtual teams could amplify this problem due to their complex characteristics and the remote location of its members.

Virtual teams have an impact on organizational structure [40,2,10] and bring unique challenges [72]. For example, virtual team members are distributed among different geographical locations and any communication and collaboration most be mediated by technology. While IT can improve efficciency, many challenges remain including time difficulties, feedback delays, misinterpretation, cultural barriers, scheduling conflicts, lack of communication, and delayed responses [25,40,3,72]. These challenges, if not addressed and managed appropriately, can affect the performance of virtual teams [69].

One such challenge is the building of trust among virtual team members. Trust is a crucial factor in forming and maintaining social relationships and is key for cooperative relationships and effective teamwork [40,86,41,72,12]. Trust includes elements of risk, vulnerability, and uncertainty which team members must overcome in order to work collaboratively [57]. In co-located organizational settings, research shows that trust has many benefits such as increasing team productivity, facilitating resolution of conflicts and disagreements, and improving effectiveness [22,33,86]. As organizations become more distributed, concerns about how to build trust among team members has grown [45]. The absence of observable behaviors, which members of traditional face-to-face teams rely upon to establish and maintain trust, makes building trust among virtual team members a complicated issue, increasing performance uncertainty [73,23].

Research needs to investigate virtual team specific factors, behaviors, and standards that will allow team members to compensate for the lack of physically observed behaviors in the establishment of trust. Kanawattanachai and Yoo [42] argue that virtual team members rely more on cognition-based trust than on affect-based trust, basing their trust decision on perceptions of evidence of trustworthiness not on genuine caring and emotion. We propose that knowledge sharing is a significant behavior which virtual team members observe, measure, and rely upon to build cognition-based trust.

3. Research model and hypotheses development

Organizations are distributed knowledge systems and the ability of the organization to identify knowledge resources,

leverage them, and make them available for its employees can lead to a distinctive competitive advantage [80,17,3]. As illustrated in our theoretical model (Fig. 1), we propose that knowledge sharing positively affects trust, collaboration and team effectiveness in virtual team settings. In addition, trust moderates the relationship between collaboration and team effectiveness, with higher levels of trust positively influencing the impact of collaboration on team effectiveness.

The Knowledge Based Theory of The Firm states that a firm is a knowledge creating entity [60,63]. Grant [29] emphasizes the role of the individual within the organization in creating knowledge and argues that the role of the organization is to integrate, store, and apply the knowledge created by the individuals. Organizations assemble virtual teams of knowledgeable and skilled individuals who are expected to utilize their knowledge to perform organizational tasks. Teams perform better when they contain members with knowledge relevant to the task at hand [55,26,64]. As with traditional teams, virtual teams perform better and produce more effective outcomes when they are able to locate, access, and apply organizational knowledge [15,26]. In a virtual setting, the knowledge must be exchanged and shared amongst team members through the use of connecting information technologies given location diffferences.

This knowledge exchange can be viewed through the lens of Social Exchange Theory where human behavior is seen as taking place in a social exchange [9]. Individuals within a social system exchange favors with a general expectation of some future, but unclear, return. Long-term relationships obtain where individuals have enough time to collaborate and exchange favors [9,58]. With these assumptions, knowledge sharing can be seen as a form of generalized social exchange, where individuals share their knowledge without any clear expectations of a return other than the promise of a long-term mutual relationship [24].

Social Exchange Theory posits that people behave in ways that maximize their benefits and minimize their costs [58]. The main cost of sharing knowledge, especially in a virtual team setting, is the loss of personal relative advantage from the release of unique knowledge. The main benefit of knowledge sharing is effective collaboration and integration of diverse resources to reach new insights. To motivate proper behavior, organizations are implementing reward systems to encourage team members to contribute and share knowledge while in some organizations, punishing employees if they refuse to share the knowledge they possess [7].

To collaborate effectively, virtual teams require distributed knowledge to be adequately shared and integrated by team members [85,70,64]. Otherwise, virtual teams will be less efficient, suffering higher costs associated with infformation search, communication failure, information misunderstanding and misinterpretation, and inadequate decision making due to missing data [30,70]. Knowledge sharing by member experts enables effective team outcomes by ensurring all pieces of an information puzzle are available for task performance and appropriate decision making. This allows the team, regardless of location, to accomplish its work requirements and contibute to an organization's goals. Thus, we hypothesize:

H1. In virtual team settings, knowledge sharing has a positive influence on collaboration.

Virtual team characteristics influence the way team members work together and have the potential to hinder team success and effectiveness. For example, without the benefit of face-to-face interaction, virtual teams tend to take longer to reach common ground and to collaborate effectively [38,71]; DeOrtentiis et al. [18].

However, virtual teams are able to overcome such challenges. As suggested by Interdependence theory, individuals within a team will naturally work collaboratively in order to achieve a common goal, ultimately affecting team effectiveness [18,44]. For example, virtual team members tend to express their opinions more freely and openly regardless of any social or managerial constrains [21]. Consequently, virtual team members are able to assess each other more accurately based on performance and contribution. Virtual team members also show less bias compared to those in traditional teams when evaluating each other's performance and contribution [84].

The result is a collaborative team effort where, despite having to overcome challenges associated with disparate locations, virtual teams exchange more ideas, share more knowledge, and effectively coordinate tasks among each other [10,56]. This tighter integration, enabled by modern technologies, affects team outcomes. We thus hypothesize:

H2. In virtual teams, collaboration among team members has a positive influence on team effectiveness.

The type of trust which develops in virtual settings is cognitive-based trust [42]. This is because: 1) information technologies are not successful in transferring feelings and emotions which affect based trust depends upon; and 2) trust decisions are often built based on team members ability, integrity, and benevolence which, requires virtual team members to provide solid evidence of their trustworthiness before other team members will trust them [57,42].

Forming trust in new virtual teams is difficult as members may lack a history of working together [40]. Especially as virtual teams are located in separate sites, lack of face-to-face interaction, work in different time zones, etc., reducing key observable cues available to traditional teams in the evaluation of team member trust [42,70]. With the lack of physical interaction which takes place in collocated team settings, virtual team members need to demonstrate different and unique behaviors to their teammates in order to prove their benevolence and integrity. Sharing knowledge is one such behavior. By sharing the knowledge they possess virtual team

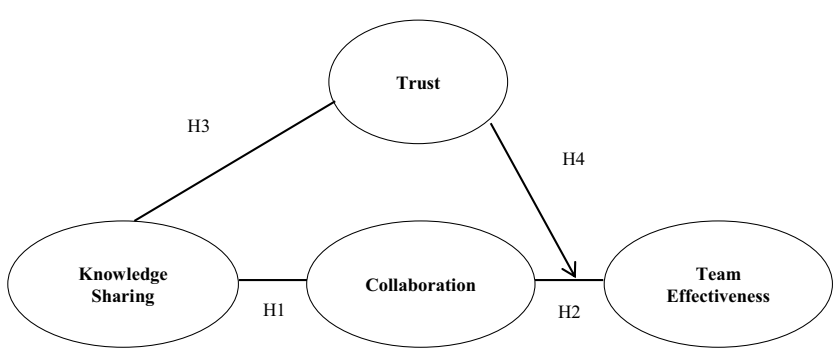


Fig. 1. Research model.

members demonstrate their willingness to act with the group in mind, that they are dependable, and can be relied upon to contribute to the collective task. Knowledge sharing provides evidence of benevolence and trustworthiness of team members. Thus, we hypothesize:

H3. In virtual team settings, knowledge sharing has a positive influence on trust among team members.

A virtual team is a social system of individuals who are expected to collaborate on a common organizational task, where the act of one team member affects all others on the team [37]. Social Capital Theory explains how changes in relations among individuals in a social system facilitate action, coordination, collaboration, and resource exchange [16,1,14]. In the social capital process, as Coleman describes,

"*If A does something for B and trusts B to reciprocate in the future, this establishes an expectation in A and obligation on the part of B. This obligation can be conceived as a credit slip held by A for performance by B···?Unless the placement of trust has been unwise and these are bad debts that will not be repaid*" [16].

Coleman's [16] argument suggests that in order to build social capital in social systems, two components are necessary: action and trust among the individuals. For each action that takes place, a trust decision is made, whether the action confirms that trust is warranted or suggests that trust is misplaced. In social systems like teams, trust is reported as a key factor in reducing risk, complexity, and uncertainty, enabling a positive atmosphere of collaboration among individuals within the system [47,40,8,66].

The role of trust in an organizational setting can be modeled in two ways [20]. The most dominant model depicts trust as having a direct effect on team outcomes [66,67,59,64], with the second depicting trust as having a moderating effect on team outcomes [41,12]. Conducting a comprehensive review of research utilizing each trust model, [20] find research utilizing model 1 (trust as a main effect) has inconsistent, non-robust results, especially when trust is hypothesized to have positive effects on workplace behaviors and performance outcomes [86,19]. On the other hand, research adopting model 2 (trust as a moderator) demonstrates that trust provides a necessary condition for better team outcomes and reveals more consistent findings [90,20]. As highlighted in Fig. 1, we adopt model 2 propose that trust as a moderator of the collaboration-effectiveness relationship.

Team collaboration is the backbone which supports and drives team success and effectiveness. However, the ability to collaborate depends heavily upon trust to facilitate sharing of information and knowledge across the team [77,66,46]. Without the appropriate information and know how, individuals and teams cannot make effective decisions. Moreover, without trust, members may exert a less than optimal collaborative effort towards the team outcome. A team will still produce output, however when trust levels among members are low, virtual team production may be minimized [6,41,64]. Thus, we hypothesize:

H4. (a) In high trust contexts, there will be a positive association between collaboration and team effectiveness. (b) In low trust contexts, the association between collaboration and team effectiveness will be significantly less strong.

4. Methodology

In this quantitative, positivist research, we test our proposed research model analyzing primary survey data. To avoid sampling bias, we focused on a specific industry, administering a questionnaire to virtual team members who work in an organization setting, in the information technology industry (e.g. software engineers, software developers, etc.). To acquire a representative sample, the sample frame in this study was acquired from LinkedIn.com, a leading professional social network.

The data was collected in two phases. In the first phase, we identified individuals and groups who best represented our population of interest, targeting those profiles containing a major IT firm's name (e.g. Microsoft, Google, IBM . . . etc.). This is possible as in LinkedIn, members post detailed resumes which include education, past experience, and current employment information. LinkedIn also provides suggestions of other people who work for the same company, affiliated companies, and similar companies. These features enabled us to target appropriate subjects for our study. Emails were then sent to both group administrators and directly to group members. A brief description of the research was given along with a link to access the survey online. Over 1000 invitations to participate were sent. A total of 193 subjects completed the survey, representing an approximate 19.3% response rate. However, as group administrators were encouraged to send the invitation on their team members, a true response rate could not be calculated.

The survey items were derived from previously published research. The variables of interest are knowledge sharing, trust, collaboration, and team effectiveness. The measures for knowledge sharing is adopted from Phang et al. [68]. The measures for trust were derived from Mayer et al. [57], a widely cited and reference research project on the topic (e.g. Jarvenpaa et al. [40]). Measures for collaboration were derived from Aram and Morgan [5], while measures for team effectiveness were derived from Lurey and Raisinghani [54]. The latter measures virtual team effectiveness in terms of team performance and team member's satisfaction. Our survey questions are shown in Appendix A.

5. Data analysis

A demographic and descriptive analysis of the data is shown in Table 2. The structural equation model is evaluated using the partial least squares (PLS) method. PLS is recommended for complex models focusing on prediction, and allows for minimal demands on measurement scales, sample size, and residual distribution [13]. Reliability is tested using Cronbach's alpha and Dillon-Goldstein's rho, as proposed by Chin [13]. As discussed below, the validity of the model is assessed by evaluating content validity, convergent validity, and discriminant validity using generally accepted tests. Finally, the Sobel test of mediation is used along with control variable analysis and multi-group analysis.

5.1. Demographics and descriptive statistics

Respondents were asked to indicate their gender and age, and to answer a number of questions concerning factors that potentially could impact team effectiveness. These factors include how long an individual has (1) worked in virtual team settings, (2) been members of the same virtual team, (3) participated in pure virtual environment or in both virtual and face-to-face environments, (4) been members of global virtual teams, (5) been a virtual team leader, (6) worked for the same organization, and (7) if sharing knowledge is mandatory or voluntarily (Table 1). A Multi-group analysis is conducted to verify if any of these factors have an effect.

5.2. Reliability and validity

The Partial Least Squares (PLS) method performs a two-stage analysis using confirmatory factor analysis to assess the measurement model followed by an examination of the structural relationships. Tables 1 and 3 display the measurement model

Table 1
Demographics and descriptive statistics.

	Category	Percentage of Participants	Mean	Standard Deviation	Sample Variance
Gender	Male	77%	1.23	0.42	0.18
	Female	23%			
Age	1. 18–20	0	2.81	0.88	0.88
	2. 21–29	43%			
	3. 30–39	40%			
	4. 41–50	12%			
	5. 51–60	5%			
Experience	1. less than a year	11%	2.27	0.74	0.77
	2. 1–5	53%			
	3. 6–10	27%			
	4. 10–15	6%			
	5. more than 15	3%			

	Category	Percentage of Participants	Mean	Standard Deviation	Sample Variance
Member of Current Team	1. less than a year	28%	2.27	0.74	0.55
	2. 1–5	54%			
	3. 6–10	12%			
	4. 10–15	6%			
	5. more than 15	0%			
Virtual Team Type	1. Online only	6%	1.85	0.65	0.42
	2. Combined online and face-to-face members	35%			
	3. Both Types	59%			
Leader vs. Member	1. Leader	42%	0.42	0.49	0.24
	2. Member	58%			
Global Virtual Team Member	1. Yes	60%	0.42	0.49	0.24
	2. No	40%			
Members work for the same organization	1. Yes	82%	0.60	0.49	0.24
	2. No	18%			
Mandatory vs. Voluntarily	1. Yes	76%	0.82	0.39	0.15
	2. No	24%			

analysis results. Reliability of the measurement model is assessed by examining internal consistency and indicator reliability. Internal consistency measures the reliability of a set of indicators, represented by Cronbach's alpha and Dillon-Goldstein's rho (Table 2). Validity of the measurement model is assessed by examining content validity, internal consistency and discriminant validity. Content validity is ensured by utilizing measurement items validated in existing research.

The psychometric properties of the research model were evaluated by examining item loadings, internal consistency, and discriminant validity. Research suggests that item loadings and internal consistencies greater than 0.70 are considered acceptable [32]. As can be seen in Table 2, all scores surpass this threshold. Internal consistency is evaluated by a construct's composite reliability score (Table 3) which shows adequate reliability scores for all constructs.

Evaluating discriminant validity has two parts; firstly, each item should load higher on its respective construct than on the other constructs in the model, and secondly, the Average Variance Extracted (AVE) for each construct should be higher than the inter-construct correlations. The analysis shows that all items load higher on their respective construct than the other constructs in the research model. Likewise, the square root of the AVE in the diagonal for each construct is higher than the inter-construct correlations on the same row and the same column as illustrated in Table 4. This suggests that the model has good discriminant validity.

Table 2
Reliability Dillon-Goldstein's rho.

Construct	Cronbach's Alpha	
Knowledge Sharing	0.93	0.95
Trust	0.83	0.89
Collaboration	0.91	0.94
Team Effectiveness	0.92	0.95
Communication	0.90	0.93
Coordination	0.91	0.94
Performance	0.91	0.96
Satisfaction	0.92	0.94

5.3. Structural model

The structural model evaluation results are presented in Fig. 2. Knowledge Sharing has no R-squared value as it is an exogenous variable. Our model explains 31% of the variance in Trust, 49% of collaboration variance, and 50% of the variance in team effectiveness. The path coefficients between knowledge sharing and trust, knowledge sharing and collaboration, and collaboration and team effectiveness are significant at 0.001. This indicates support for hypotheses H1-H3. However, the path coefficient for the moderating effect of trust on the relationship between collaboration and

Table 3
Composite Reliability Scores.

Construct	CR
Knowledge Sharing	0.95
Trust	0.85
Collaboration	0.95
Effectiveness	0.96

Table 4
Discriminant validity.

Construct	AVE	Knowledge Sharing	Trust	Collaboration	Effectiveness
Knowledge Sharing	0.84	**0.92**			
Trust	0.88	0.65	**0.94**		
Collaboration	0.71	0.70	0.61	**0.84**	
Effectiveness	0.75	0.64	0.43	0.71	**0.87**

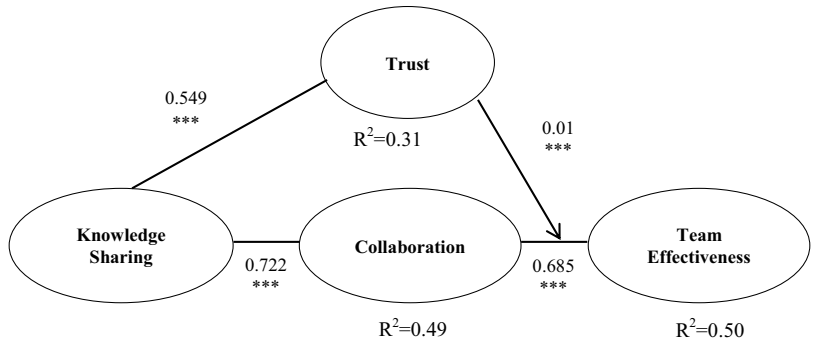


Fig. 2. PLS SEM results.

team effectiveness is not supported (H4). Table 5 summaries these conclusions.

5.4. Mediation analysis

Sobel Test for the Significance of Mediation is used to test for the significance of collaboration mediating the relationship between knowledge sharing and team effectiveness. As shown in Table 6 the Sobel test for mediation is significant at the 0.01 level, which indicates that collaboration mediates the relationship between knowledge sharing and team effectiveness.

5.5. Control variable analysis

In this study, we controlled for the effect of the team design process on the dependent variables (i.e. trust and team effectiveness). Different organizations have different methods to establish virtual teams. For instance, some organizations assign members to virtual teams without asking members for their opinions, while other organizations consult with members, consider their feedback, and give them the choice to join the team they prefer. Conducting PLS analysis on the model after adding the control variable resulted in minor changes in the R-Squared values for trust and team effectiveness as shown in Table 7. Also as shown in Fig. 3, the path coefficient between the control variable and both trust and team effectiveness are not strong.

5.6. Multi-group analysis

Multi-group analysis is conducted to determine if there is any significant effect for specific individual team characteristics on the knowledge sharing behavior. Conducting a multi group analysis is important in this research due to the characteristics of virtual teams. The results indicate that the variable, being member of a global virtual team, has a significant influence for trust. This result provides an indication that the diversity of virtual team member cultures and backgrounds affects the level of knowledge shared between team members and the level of trust among them (see Tables 8 and 9).

Table 5
Summary of hypotheses tests.

Hypotheses	Supported
H1: Knowledge Sharing → Collaboration	Yes
H2: Collaboration → Team Effectiveness	Yes
H3: Knowledge Sharing → Trust	Yes
H4: Trust x Collaboration → Team Effectiveness	No

Table 6
Sobel test for the significance of mediation.

	Path Coefficient	Standard Error
Knowledge Sharing → Collaboration	0.722	0.0536
Collaboration → Team Effectiveness	0.685	0.2231
Sobel Test Statistics	2.99358902	
One-tailed probability	0.00137859	
Two-tailed probability	0.00275717	

5.7. Post-hoc analysis

The structural model analysis results suggest that trust does not have a significant moderating impact on the relationship between collaboration and team effectiveness (H4 rejected). This is in contrast to prior literature which highlights trust as having a moderating effect (e.g. [20]). To examine the role of trust further, we conducted a series of post hoc analyses that examined the different ways trust might influence overall team effectiveness. We conducted three different PLS analyses on adjusted models; in the first analysis we tested for a direct effect of trust on collaboration (Fig. 4), in the second analysis we tested for a direct effect of trust on virtual team effectiveness (Fig. 5), and in the third analysis we tested for a direct effect of trust on both collaboration and team effectiveness at the same time (Fig. 6).

The results of these analyses suggest that a statistically significant direct effect of trust on collaboration, as seen in both Figs. 4 and 6. However, the relationship between trust and team effectiveness is not statistically significant in either analysis that examined the direct impact of trust on team effectiveness, see Figs. 5 and 6.

Comparing the model shown in Fig. 4 with the original model shown in Fig. 2, we can see the revised model explains more of the

Table 7
Control Variable Analysis.

Construct	R-Squared: No Control	R-Squared: With Control	Change	Sig.
Trust	0.31	0.314	<0.1	–
Team Effectiveness	0.50	0.593	<0.1	**

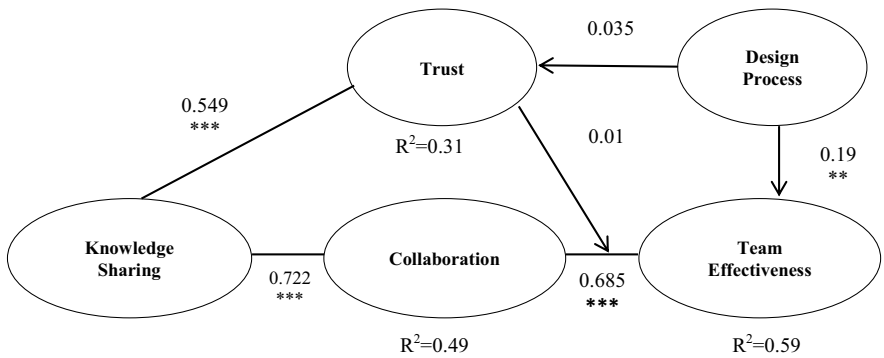


Fig. 3. Control variable analysis.

Table 8
Multi-group moderating effect (Trust).

Variable	t-statistics	p-value	Significance
Experience	1.22	0.22	Not Significant
Member of the current virtual team	0.594	0.554	Not Significant
Leader vs. Member	0.078	0.938	Not Significant
Global virtual team	2.238	0.027	**Significant at 0.05**
Mandatory or voluntarily	1.162	0.247	Not Significant
Meet in Person	0.848	0.398	Not Significant

Table 9
Multi-multigroup moderating effect (Collaboration).

Variable	t-statistics	p-value	Significance
Experience	0.071	0.944	Not Significant
Member of the current virtual team	0.628	0.531	Not Significant
Leader vs. Member	0.164	0.870	Not Significant
Global virtual team	1.067	0.287	Not Significant
Mandatory or voluntarily	1.162	0.247	Not Significant
Meet in Person	1.191	0.235	Not Significant

variation in team effectiveness than the originally hypothesized model. Thus, the results of the post hoc analysis confirm that trust does play an important role in the overall effectiveness of a team. However, it does not have a direct effect on team effectiveness, instead the impact of trust on team effectiveness is fully mediated by collaboration.

6. Discussion

Our results have significant implications for both research and practice. For research, it provides new insights regarding the role of knowledge sharing in virtual teams. We also identify the need for future research to better understand the impact of trust on virtual teams' outcomes. For practice, this research emphasizes the significance of knowledge sharing to virtual team effectiveness, and recommends that organizations support knowledge sharing in virtual settings on both technological and social levels.

This research investigated the relationship between knowledge sharing and trust, and how this relationship ultimately affects virtual team effectiveness. The researchers utilized the Knowledge-Based Theory of the Firm and Social Exchange Theory to frame hypotheses. Results provide support for the theoretical model and qualified support for most all but one hypothesized relationship. The results suggest that knowledge sharing has a significant influence on both trust and collaboration in virtual team settings. This provides support for hypotheses H1 and H3. Knowledge sharing in virtual team settings could be a significant element in establishing social capital and social exchange among virtual team members. Consistent with the Knowledge-Based Theory of the Firm and Nonaka's [61] concept of "ba", a virtual

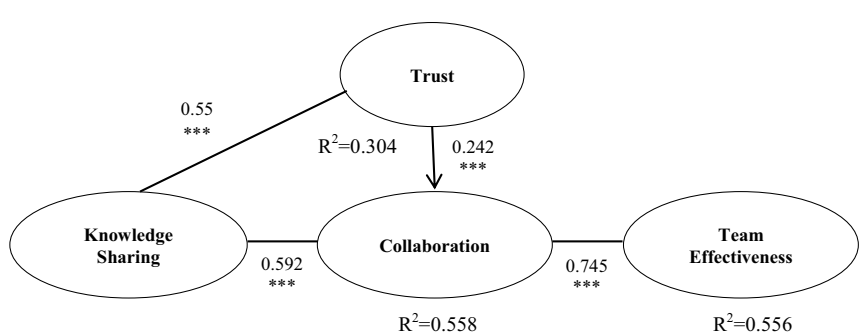


Fig. 4. Post-hoc analysis (Trust → Collaboration).

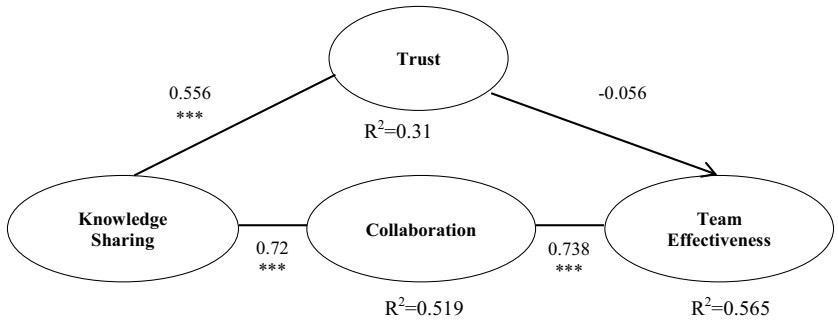
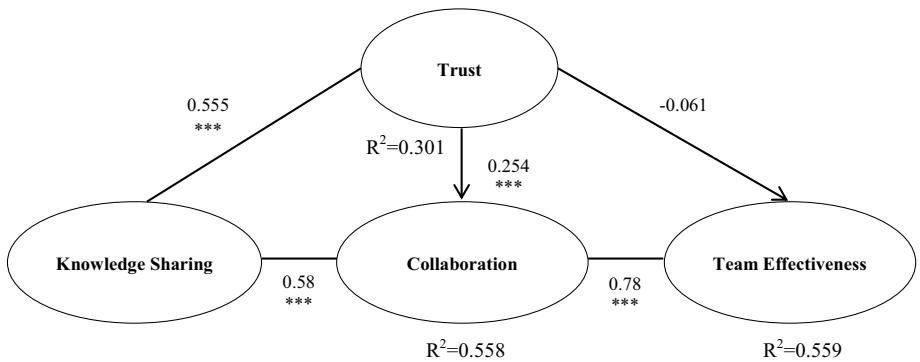


Fig. 5. Post-hoc analysis (Trust → Team Effectiveness).



team is considered a place where members share knowledge and transform their environment to reach new insights. Furthermore, this research extends the literature on virtual teams to argue that sharing knowledge is crucial for virtual team members to collaborate, trust each other, and to be effective.

The results illustrate that the relationship between collaboration and team effectiveness is significant, supporting hypothesis H2. The Sobel test of mediation indicates that collaboration mediates the relationship between knowledge sharing and team effectiveness. This provides additional support for the significance of knowledge sharing to the effectiveness of virtual teams.

Contrary to our expectations, hypothesis H4 was not supported, indicating that trust does not have a significant moderating impact on the relationship between collaboration and team effectiveness. This is consistent with contradictory and inconclusive results of related research [20,18]. Aubert and Kelsey [6] find that the level of trust among virtual team members does not have a significant direct impact on team performance. This is consistent with our results and suggests that the impact of trust may have a limited direct influence on virtual team effectiveness. However, the post-hoc analysis suggests that trust does have a direct effect on collaboration which in turn impacts virtual team effectiveness. [41] argue that trust effects are sensitive to the context of the virtual team, which might suggest that our results could be limited to the population of our study. Nevertheless, trust remains a complex phenomenon and the characteristics of virtual teams add to this complexity.

Peters and Manz [67] argue that building trust among virtual team members depends on two factors, the first factor is personality-based trust and the second factor is institution-based trust. The latter allows team members to trust each other's behavior based on the norms and rules of the organization. The findings of [64] suggest that institution-based trust affects virtual team effectiveness. Institution-based trust was the center of discussions the researchers had with several virtual team members who responded to the study's survey, in order to understand why hypothesis H4 was not supported. Those respondents argued that trust among virtual team members is not built based on personal factors; mostly virtual team members trust each other's capabilities and competencies to perform the organizational task at-hand. A virtual team member for over 10 years stated that in his organization, a team leader uses a system which matches employees' profiles to the objectives of the project. The leader then interviews candidates to decide whom should be assigned. These profiles are made available for team members before they begin working together, which means that team members establish institution-based trust at a very early stage of the project life. As a whole, future research is needed to investigate the impact of trust on virtual teams' collaboration and effectiveness in order to reach conclusive findings.

The multi-group moderation effect results were negative for all items except for global virtual team members' perception of trust. This result may indicate that trust is higher among virtual team members who have similar characteristics (i.e. culture, language). The impact of culture on trust in virtual teams has been examined in the literature given that team members of different cultures are likely to join the same virtual team [87,36,82]. Culture is fundamental in interpreting communication and behavior [87], which explains why having members of different cultures in the same virtual team could affect the level of trust among them. Future studies are needed to investigate ways to enhance trust among global virtual team members and how mechanisms to enhance trust may differ for global teams when compared to teams formed from a single country.

7. Limitations

Our research investigates knowledge sharing in teams independent of technology and task. This is a potential limitation as the

type of technology used may affect the formation of trust and the knowledge shared by team members. For example, knowledge management systems (KMS) used by organizations can vary from simple blogs and discussion boards, to more sophisticated software applications designed especially for organizational knowledge needs. KMS related research suggests that these systems can indeed influence the quality and quantity of knowledge shared [2,43]. In addition, research has shown there is a relationship between task complexity and the type of technology needed to facilitate virtual team collaboration; higher task complexity requires more synchronous, tightly coupled interaction than low task complexity [74]. Future research could include an examination of different types of technologies used by teams, like KMSs, as well as the complexity of virtual team tasks to better understand the interplay between task-technology fit and the social aspect of sharing knowledge in virtual teams.

Another limitation of our research concerns the measure used for virtual team collaboration. It is possible that the measure, originally designed for co-located team settings [5], may affect the point estimates of our results. It is possible that the nuances of online collaboration may lead to differences in team collaboration than what takes place in face-to-face settings. For example, prior research has found that there is a need for technical training to empower team members to function in virtual environments, in addition to traditional training on developing good interpersonal skills [45]. Future research could develop and use measures specifically designed for virtual team settings to provide additional insights into the balance and influence of technical and interpersonal skills on virtual team collaboration.

Finally, our study utilized virtual team members who worked in the information technology industry as the source of our data measures. As technical skills have been found to be an important factor for virtual team success (e.g. Kirkman et al. [45]), future research should examine data samples from virtual team members in non-technical industries in order to improve the generalizability of our results. It is possible that different backgrounds may affect the estimated collaborative relationships.

8. Conclusion

Organizations are increasingly using virtual teams to take advantage of employee skills and knowledge regardless of geographical location. Yet, little is known about how sharing knowledge impacts the relationship between virtual team members, their collaboration, and team effectiveness. This research proposed a conceptual model which hypothesized a relationship between knowledge sharing, trust, collaboration, and team effectiveness in virtual team settings. The theoretical model was developed, based on the Knowledge Based Theory of The Firm, Social Capital Theory, and the Social Exchange Theory. PLS analysis results support three hypotheses, indicating that knowledge sharing influences the formation of trust within virtual teams and contributes to virtual team effectiveness. The study finds the relationship between knowledge sharing and team effectiveness is mediated by collaboration.

The study raises important questions for future research. First, it is unclear why trust does not seem to directly influence team effectiveness. Instead the results of the post hoc analysis suggest that trust has a direct impact on collaboration and it is that collaboration that impacts team effectiveness. Prior research has suggested that the impact of trust in virtual teams is sensitive to the virtual team setting. Additional research is necessary to determine if the importance of trust on team effectiveness is higher in other virtual team environments (e.g. for online communities of practice outside the organization). Future research should also examine the impact of different team building strategies on trust, collaboration and team effectiveness, especially in global virtual team contexts.

Appendix A.

Demographics

Please indicate whether you are	o Male				o Female	
What is your age group?	o 18-20	o 21-29	o 30-39	o 40-49	o 50-59	o 60 or above
How long have you worked as a virtual team member? (In Years)	o < 1 year	o 2-5	o 6-10	o 11-15	o more than 15	
Are you currently a virtual team member?	o Yes				o No	
Have you ever been a virtual team leader?	o Yes				o No	
Have you ever been part of a global virtual team?	o Yes				o No	
Do all members of your team work for the same organization?	o Yes				o No	
Is it mandatory to share knowledge in your organization?	o Yes				o No	
Have you participated in a purely online virtual team or in a team that combined online and face-to-face work? o Participated in online team(s) only o Participated in combined online and face-to-face teams o Participated in both types of virtual teams o I have never participated in a virtual team						

Survey Items

	Strongly Disagree	Disagree	Somewhat Disagree	Undecided	Somewhat Agree	Agree	Strongly Agree
Team members were asked for their suggestions when the team was originally formed	-	-	-	-	-	-	-
Careful consideration was given to the team's objectives during the design of this team.	-	-	-	-	-	-	-
I received sufficient information to understand the team's purpose when I was notified that I will be part of this team.	-	-	-	-	-	-	-
My role in the team was clearly explained to me.	-	-	-	-	-	-	-
I have received training focused on becoming more effective in the virtual team setting.	-	-	-	-	-	-	-
My virtual team is equipped with adequate tools and technologies to perform our tasks.	-	-	-	-	-	-	-
The electronic methods we use to communicate with one another are effective.	-	-	-	-	-	-	-
My team members were given the opportunity to meet each other in person.	-	-	-	-	-	-	-
I'm rewarded individually for my work effort	-	-	-	-	-	-	-
I routinely share my knowledge with my team members	-	-	-	-	-	-	-
I routinely seek out knowledge from other team members	-	-	-	-	-	-	-
If I had my way, I wouldn't let the other team members have any influence over issues that are important to the project.	-	-	-	-	-	-	-
I really wish I had a good way to oversee the work of the other team members on the project.	-	-	-	-	-	-	-
My virtual team members communicate in a positive manner to one another.	-	-	-	-	-	-	-
If a team member makes a mistake, others generally point out his error and discuss it with him.	-	-	-	-	-	-	-
My team has been effective in reaching its goals.	-	-	-	-	-	-	-
My team is meeting its business objectives.	-	-	-	-	-	-	-
Overall, I find the information I need to do my job.	-	-	-	-	-	-	-
I would be comfortable giving the other team members complete responsibility for the completion of this project.	-	-	-	-	-	-	-
I can turn to my team mates for help when needed.	-	-	-	-	-	-	-
I would be comfortable giving the other team members a task or problem which was critical to the project, even if I could not monitor them.	-	-	-	-	-	-	-
I try to be honest about what I think and feel when working with my team members.	-	-	-	-	-	-	-
When several team members are discussing an issue, I can ask questions about anything I do not understand.	-	-	-	-	-	-	-
My input is valued by my team members.	-	-	-	-	-	-	-

My team members and I respect each other.	-	-	-	-	-	-	-
In my team, members' morale is high.	-	-	-	-	-	-	-
My virtual team members share knowledge from work experience with each other.	-	-	-	-	-	-	-
I share my expertise from my education or training with other team members.	-	-	-	-	-	-	-
I am successful in transferring what I've learned to my team.	-	-	-	-	-	-	-
In carrying out our duties, my team members and I try to act as consultants to each other.	-	-	-	-	-	-	-
My team members and I regulate activities with each other.	-	-	-	-	-	-	-
Generally, my team is most concerned with finding the best solution.	-	-	-	-	-	-	-
My team members and I build on each other's ideas.	-	-	-	-	-	-	-
My team members and I coordinate actions and decisions well.	-	-	-	-	-	-	-
In my team, work items I depend on are changed without my knowledge.	-	-	-	-	-	-	-
My virtual team frequently faces problems deciding which member has responsibility for a work item	-	-	-	-	-	-	-
Generally, my team completes its work on time.	-	-	-	-	-	-	-
Generally, my team completes its work within the budget.	-	-	-	-	-	-	-
I enjoy being a member of this team.	-	-	-	-	-	-	-
In the future, I would be interested in participating in another virtual team.	-	-	-	-	-	-	-

References

[1] P. Adler, S. Kwon, Social capital: prospects for a new concept, Acad. Manage. Rev. 27 (1) (2002) 17–40.
[2] M. Alavi, D. Leidner, Review: knowledge management and knowledge management systems: conceptual foundations and research issues, MIS Q. 25 (1) (2001) 107–136.
[3] M. Alavi, A. Tiwana, Knowledge integration in virtual teams: the potential role of KMS, J. Am. Soc. Inf. Sci. Technol. 53 (12) (2002) 1029–1037.
[5] J. Aram, C. Morgan, The role of project team collaboration in R & D performance, Manage. Sci. 22 (10) (1976) 1127–1137.
[6] B.A. Aubert, B.L. Kelsey, Further understanding of trust and performance in virtual teams, Small Group Res. 34 (5) (2003) 575–618.
[7] K. Bartol, A. Srivastava, Encouraging knowledge sharing: the role of organizational reward systems, J. Leadersh. Organiz. Stud. 9 (1) (2002) 64–76.
[8] M. Baba, Dangerous liaisons: trust, distrust, and information technology in American work organizations, Hum. Organ. 58 (3) (1999) 331–346.
[9] P. Blau, Power and Exchange in Social Life, Wiley, New York, 1964.
[10] G.R. Berry, Enhancing effectiveness on virtual teams: understanding why traditional team skills are insufficient, J. Bus. Commun. 48 (2) (2011) 186–206 (0021943610397270).
[11] G. Bock, R. Zmund, Y. Kim, J. Lee, Behavioral intention formation in knowledge sharing: examining the roles of extrinsic motivators, social-psychological forces, and organizational climate, MIS Quarterly 29 (1) (2005) 87–111 (Special Issue on Information Technologies and Knowledge Management).
[12] T. Brahm, F. Kunze, The role of trust climate in virtual teams, J. Manage. Psychol. 27 (6) (2012) 595–614.
[13] W.W. Chin, The partial least squares approach for structural equation modeling, in: G.A. Marcoulides (Ed.), Modern Methods for Business Research, Lawrence Erlbaum Associates, London, 1998, pp. 295–336.
[14] C. Chiu, M. Hsu, E. Wang, Understanding knowledge sharing in virtual communities: an integration of social capital and social cognitive theories, Decis. Support Syst. 42 (3) (2006) 1872–1888.
[15] E. Civi, Knowledge management as a competitive asset: a review, Mark. Intell. Plann. 18 (4) (2000) 166–174.
[16] J. Coleman, Social capital in the creation of human capital. american journal of sociology, vol. 94, supplement: organizations and institutions, Sociol. Econ. Approaches Anal. Soc. Struc. (1988) S95–S120.
[17] T. Davenport, L. Prusak, Working Knowledge, Harvard Business School Press, Boston, 1998.
[18] P.S. DeOrtentiis, J.K. Summers, A.P. Ammeter, C. Douglas, G.R. Ferris, Cohesion and satisfaction as mediators of the team trust–team effectiveness relationship: an interdependence theory perspective, Career Dev. Int. 18 (5) (2013) 521–543.
[19] K.T. Dirks, The effects of interpersonal trust on work group performance, J. Appl. Psychol. 84 (1999) 445–455.
[20] K. Dirks, D. Ferrin, The role of trust in organizational settings, Organiz. Sci. 12 (4) (2001) 450–467.
[21] D. Duarte, N. Snyder, Mastering Virtual Teams, Jossey-Bass, San Francisco, 1999.
[22] P. Earley, Trust, perceived importance of praise and criticism, and work performance: an examination of feedback in the United States and England, J. Manage. 12 (1986) 457–473.
[23] P.G.S. Ferreira, E.P. de Lima, S.E.G. da Costa, Developing a methodology for assessing virtual teams' performance perception, Int. J. Product. Perform. Manage. 61 (7) (2012) 710–729.
[24] J. Fulk, A.J. Flanagin, M.E. Kalman, P.R. Monge, T. Ryan, Connective and communal public goods in interactive communication systems, Commun. Theory 6 (1) (1996) 60–87.
[25] S. Fussell, R. Kraut, J. Lerch, W. Scherlis, M. McNally, Cadiz, Coordination, overload and team performance: effects of team communication strategies, Proceedings of The1998 ACM Conference on Computer Supported Cooperative Work, NY, US., 1998.
[26] H. Gardner, F. Gino, B. Staats, Dynamically integrating knowledge in teams: transforming resources into performance, Acad. Manage. J. 55 (4) (2012) 998–1022.
[27] D. Gilmour, How to fix knowledge management, Harv. Bus. Rev. 10 (2003) 16–17.
[28] A.H. Gold, A. Malhotra, A.H. Segars, Knowledge management: an organizational capabilities perspective, J. Manage. Inf. Syst. 18 (1) (2001) 185–214.
[29] R. Grant, Prospering in dynamically-competitive environments: organizational capacity as knowledge integration, Organiz. Sci. 7 (4) (1996) 375–387.
[30] P. Gray, The impact of knowledge repositories on power and control in the workplace, Inf. Technol. People 14 (4) (2001) 368–384.
[31] T. Griffith, J. Sawyer, M. Neale, Virtualness and knowledge in teams: managing the love triangle of organizations, individuals, and information technology, MIS Q. 27 (2) (2003) 265–287.

[32] F. Hair, E. Anderson, L. Tatham, C. Black, Multivariate Data Analysis, 6th ed., Prentice-Hall, Upper Saddle River, New Jersey, 2005.
[33] J.M. Hagen, S. Choe, Trust in Japanese interfirm relations: institutional sanctions matter, Acad. Manage. Rev. 23 (3) (1998) 589–600.
[35] A. Hemetsberger, C. Reinhardt, Collective development in open-source communities: an activity theoretical perspective on successful online collaboration, Organiz. Stud. 30 (9) (2009) 987–1008.
[36] K. Henttonen, K. Blomqvist, Managing distance in a global virtual team: the evolution of trust through technology-mediated relational communication, Strateg. Change 14 (2) (2005) 107–119.
[37] M. Hoegl, H. Gemuenden, Teamwork quality and the success of innovative projects: a theoretical concept and empirical evidence, Organiz. Sci. 12 (4) (2001) 435–449.
[38] J. Holton, Building trust and collaboration in a virtual team, Team Perform. Manage. 7 (3/4) (2001) 36–47.
[40] S. Jarvenpaa, D. Leidner, Communication and Trust in Global Virtual Teams, 10, Organization Science, 1999, pp. 791–815 No. 6.
[41] S. Jarvenpaa, T. Shaw, D. Staples, Toward a contextualized theories of trust: the role of trust in global virtual teams, Inf. Syst. Res. 15 (3) (2004) 250–267.
[42] P. Kanawattanachai, Y. Yoo, Dynamic nature of trust in virtual teams, J. Strateg. Inf. Syst. 11 (3–4) (2002) 187–213.
[43] A. Kankanhalli, B. Tan, K. Wei, Contributing knowledge to electronic knowledge repositories: an empirical investigation, MIS Quarterly 29 (1) (2005) 113–114.
[44] H.H. Kelley, J.W. Thibaut, Interpersonal Relations: A Theory of Interdependence, Wiley, New York, 1978, pp. 341.
[45] B. Kirkman, B. Rosen, B. Gibson, P. Tesluk, McPherson, O. Simon, Five challenges to virtual team success: lessons from Sabre, Inc, Acad. Manage. Executive 16 (3) (2002) 67.
[46] B. Koehne, P. Shih, J. Olson, Remote and alone: coping with being the remote member on the team, Proceedings of the ACM 2012 Conference on Computer Supported Cooperative Work (CSCW '12) (2012).
[47] P. Kollock, The emergence of exchange structures: an experimental study of uncertainty, commitment, and trust, Am. J. Sociol. (1994) 313–345.
[51] A. Lam, Tacit knowledge, organizational learning and societal institutions: an integrated framework, Organiz. Stud. 21 (3) (2000) 487–513.
[54] J. Lurey, M. Raisinghani, An empirical study of best practices in virtual teams, J. Inf. Manage. 38 (2001) 523–544.
[55] A. Malhotra, A. Majchrzak, Enabling knowledge creation in far-flung teams: best practices for IT support and knowledge sharing, J. Knowl. Manage. 8 (4) (2004) 75–88.
[56] J. Mathieu, M. Maynard, T. Rapp, L. Gilson, Team effectiveness 1997–2007: A review of recent advancements and a glimpse into the future Journal of Management, J. Manage. 34 (3) (2008) 410–476.
[57] R.C. Mayer, J.H. Davis, F.D. Schoorman, An integrative model of organizational trust, Acad. Manage. Rev. 20 (3) (1995).
[58] L. Molm, N. Takahashi, G. Peterson, Risk and trust in social exchange: an experimental test of a classical proposition, Am. J. Sociol. 105 (5) (2000) 1396–1427.
[59] M. Muethel, F. Siebdrat, M. Hoegl, When do we really need interpersonal trust in globally dispersed new product development teams? R&D Manage. 42 (1) (2012) 31–46.
[60] I. Nonaka, A dynamic theory of organizational knowledge creation, Organiz. Scie. 5 (1994) 14–37.
[61] I. Nonaka, H. Takeuchi, The Knowledge Creating Company, Oxford University Press, New York, 1995.
[63] I. Nonaka, R. Toyama, A. Nagata, A firm as a knowledge-creating entity: a new perspective on the theory of the firm, Ind. Corp. Change 9 (1) (2000) 1–20.
[64] F. Pangil, J. Moi Chan, The mediating effect of knowledge sharing on the relationship between trust and virtual team effectiveness, J. Knowl. Manage. 18 (1) (2014) 92–106.
[66] D. Paul, R.R. McDaniel, Effect of interpersonal trust on virtual collaborative relationship performance, MIS Q. 28 (2) (2004) 183–227.
[67] L. Peters, C. Manz, Identifying antecedents of virtual team collaboration, Team Perform. Manage. 13 (3/4) (2007) 117–129.
[68] C. Phang, A. Kankanhalli, R. Sabherwal, Usability and sociability in online communities: a comparative study of knowledge seeking and contribution, J. Assoc. Info. Syst. 10 (2009) 721–747.
[69] G. Piccoli, A. Powell, B. Ives, Virtual teams: team control structure, work processes, and team effectiveness, J. Info. Technol. People 17 (4) (2004).
[70] P. Pinjani, P. Palvia, Trust and knowledge sharing in diverse global virtual teams, Info. Manage. 50 (4) (2013) 144–153.
[71] R. Potter, P. Balthazard, Understanding human interaction and performance in the virtual team, J. Info. Technol. Theory Appl. (JITTA) 4 (1) (2002) 1–23.
[72] A. Powell, G. Piccoli, B. Ives, Virtual teams: a review of current literature and directions for future research, Data Base Adv. Inf. Syst. 35 (1) (2004).
[73] J. Riegelsberger, A. Sasse, J. McCarthy, The researcher's dilemma: evaluating trust in computer-mediated communication, Int. J. Hum. Comput. Stud. 58 (6) (2003) 759–781.
[74] K. Riopelle, J.C. Gluesing, T.C. Alcordo, M. Baba, D. Britt, W. McKether, L. Monplaisir, H.H. Ratner, K.H. Wagner, Context, Task, and the Evolution of Technology Use in Global Virtual Teams. Virtual Teams That Work: Creating Conditions for Virtual Team Effectiveness, John Wiley & Sons, San Francisco, CA, 2003, pp. 239–264.
[75] A. Cheng, Communication and Conflict in Virtual Teams, ProQuest, 2008.
[77] J.E. Scott, Facilitating interorganizational learning with information technology, J. Manage. Inf. Syst. 17 (2) (2000) 81–113.
[78] K. Squire, C. Johnson, Supporting distributed communities of practice with interactive television, Educ. Technol. Res. Dev. 48 (1) (2000) 23–43.
[79] S.I. Tannenbaum, J.E. Mathieu, E. Salas, D. Cohen, Teams are changing: are research and practice evolving fast enough? Ind. Organiz. Psychol.: Perspect. Sci. Pract. 5 (1) (2012) 2–24.
[80] H. Tsoukas, The firm as a distributed knowledge system: a constructionist approach, Strateg. Manage. J. 17 (1996) 11–25.
[81] F. Van den Bosch, H. Volberda, M. Boer, Co-evolution of firm absorptive capacity and knowledge environment: organizational forms and combinative capabilities, Organiz. Sci. 10 (5) (1999) 551–568.
[82] R.M. Verburg, P. Bosch-Sijtsema, M. Vartiainen, Getting it done: critical success factors for project managers in virtual work settings, Int. J. Project Manage. 31 (1) (2013) 68–79.
[84] S. Weisband, L. Atwater, Evaluating self and others in electronic and face-to-face groups, J. Appl. Psychol. 84 (1999) 632–639.
[85] Y.C. Xiao, Y.H. Jin, The hierarchical linear modeling of shared mental model on virtual team effectiveness, Kybernetes 39 (8) (2010) 1322–1329.
[86] Zaheer, V. Perrone, B. McEvily, Does trust matter? Exploring the effects of inter-organizational and interpersonal trust on performance, Organiz. Sci. 9 (2) (1998) 141–159.
[87] N. Zakaria, A. Amelinckx, D. Wilemon, Working together apart? building a knowledge sharing culture for global virtual teams, Creativity Innov. Manage. 13 (1) (2004) 15–29.
[88] R. Zolin, P.J. Hinds, R. Fruchter, R.E. Levitt, Interpersonal trust in cross-functional, geographically distributed work: a longitudinal study, Inf. organiz. 14 (1) (2004) 1–26.
[90] P. Hwang, W. Burgers, Properties of trust: An analytical view, Organizational Behavior and Human Decision Processes 69 (1997) 67–73.

Dr. Mohammad K. Alsharo is an Assistant Professor at Al al-Bayt University School of Information Technology. He received his PhD in Computer Science and Information Systems from the University of Colorado Denver Business School and his MS in Computer Science from the Jordan University of Science and Technology. Before his doctoral studies, he worked as a project manager. His current research focuses on virtual teams, knowledge management, and eLearning. He has published his articles in the International Journal of Emerging Technologies in Learning and in the International Journal of Business Information Systems.

Dr. Dawn G. Gregg is the Associate Dean of Programs and a Professor of Information Systems at the University of Colorado Denver Business School. She received her PhD in Computer Information Systems and her MS in Information Management from Arizona State University. Before her doctoral studies, she worked as a research and development engineer for 9 years. Her current research focuses on how to organize and maintain web-based content, so that it can be used to better meet business and special educational needs. Her articles have been published in many journals such as MIS Quarterly; the International Journal of Electronic Commerce; IEEE Transactions on Systems, Man, and Cybernetics; Communications of the ACM; and Decision Support Systems.

Dr. Ronald Ramirez is an Associate Professor at the Business School, University of Colorado Denver. His research addresses the value of technology to investing organizations, blending information technology (IT), economics, and organizational theories. His research areas of interest include IT-enabled organizational change, strategic partnerships for supply chain efficiency, firm innovation for market and revenue growth, information governance, and organizational value of healthcare IT. His articles have been published in many journals such as Information Systems Research, Decision Support Systems, Journal of Management Information Systems, and Information Systems Journal.